

Kuan-Chen Chen

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EDUCATION

University of Florida

Master of Science in Applied Data Science

Gainesville, FL

8/2024 - 5/2026

National Sun Yat-Sen University

Bachelor of Science in Applied Mathematics

Kaohsiung, Taiwan

9/2019 - 6/2023

Honors & Awards: The Valedictorian of Department of Applied Mathematics, Distinguished Service Award

EXPERIENCE

NVIDIA NeMo Automodel ([MiniCPM5-1B](#), [Falcon H1](#))

Open-Source Contributor

5/2026-Present

- Contributed 2 merged PRs to NVIDIA's distributed LLM training library, adding 7 fine-tuning recipes (SFT, LoRA) for models from 0.5B to 34B.
- Configured LoRA hyperparameters (rank, alpha, target modules) and tensor parallelism for 34B-scale training; validated convergence on T4 GPU runs.

Florida Gators Women's Basketball

Gainesville, FL

Data Scientist Intern

8/2025 - 5/2026

- Built an opponent-strength-adjusted player evaluation system (OA-Rating) using XGBoost and SHAP, analyzed 120K+ game-level observations across 5,500+ NCAA players and 350+ teams
- Implemented an Elo-based team rating system ($\rho = 0.83$ vs. NCAA NET) to track real-time team strength and support game preparation
- Developed a production R Shiny dashboard for player comparison and opponent scouting, delivering insights that contributed to team improvement from 16-17 (NET 60) to 18-15 (NET 48)

Bioinformatics Lab, University of Florida

Gainesville, FL

Student Research Assistant

12/2024 - Present

- Designed stacking ensemble algorithm combining XGBoost, LightGBM, and Balanced Random Forest with logistic regression meta-learner to classify 10,486 T-cell receptor CDR3 sequences, improving robustness on highly imbalanced data (1:23) and achieving ROC-AUC 0.876
- Developed biologically-informed data augmentation algorithm using BLOSUM62 substitution matrix, generating conservative sequence variants to expand minority class 10x (432 $\rightarrow$ 4,622) and mitigate severe class imbalance (1:23 $\rightarrow$ 1:1.5)
- Engineered 42 physicochemical features from protein sequences and optimized decision thresholds under class imbalance, improving precision to 0.716 for high-confidence, interpretable predictions

PROJECT

cAIuldron (Recipe Generation LLM) [Ckcinnabar/cAIuldron](#)

Gainesville, FL

- Built an AI recipe app that turns ingredient photos into complete recipes with AI-generated images; shipped two production versions, cutting pipeline latency from 40-80s to 10-20s
- Developed v1 pipeline using CLIP and DETR for multi-ingredient detection across 525+ food classes (88% F1); fine-tuned Llama 3.2 1B via QLoRA on 7,913 curated RecipeNLG recipes, updating only 0.23% of parameters (2.8M) while achieving 92/100 recipe quality score
- Rebuilt v2 on LangGraph with Groq-hosted Llama 4 Scout (vision) and Llama 3.3 70B (generation); parallel fork/join ChromaDB RAG + Tavily search, removing the 4-6 GB VRAM dependency

Climate Change Innovation Competition, The Ministry of Education (Taiwan)

Taipei, Taiwan

- Built DARTS campus mobility prediction system by integrating multi-source APIs (TDX/PTX), enabled automated 24/7 data pipeline and real-time forecasting of pedestrian, bicycle, and bus demand
- Developed hybrid time-series model (ARIMA-LSTM) with data preprocessing and feature engineering, achieved 86.1% accuracy; delivered actionable insights adopted by university, optimizing bus schedules and reaching national finals (Top 20 of 121 teams)

SKILLS

Languages & ML: Python, SQL, R | PyTorch, scikit-learn, XGBoost, LightGBM, SHAP

LLM & GenAI: HuggingFace Transformers, LoRA/QLoRA, LangGraph, LangSmith, ChromaDB

Web & Viz: Flask, React.js, Gradio, Dash, Shiny, Power BI, Tableau | DB: MySQL, MongoDB